

To What Extent Do Deal Premiums in M&A Transactions Reflect Synergy Expectations Versus Managerial Hubris?

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ABSTRACT

The aim is to examine the extent to which acquisition premiums paid in mergers and acquisitions (M&A) transactions across BRICS economies reflect rational synergy expectations versus behavioural distortions which arise from managerial pride. The study will draw on the ‘foundational hubris hypothesis’ advanced by Roll (1986) and the synergy taxonomy conceptualized by Bradley, Desai and Kim (1988) and integrate evidences from CEO overconfidence proxies (Malmendier & Tate, 2008). Apart from that, it will also incorporate event-study findings on acquirer-cumulative abnormal returns and country-specific evidences covering India, China, Brazil, Russia and South Africa. The analysis will prove that while synergy logic offers the most defensible ex-ante justification for premiums, the cross-sectional dispersion in long-run acquirer performance and the systematic underperformance of overconfidence-flagged bidders is consistent with a meaningful hubris contribution, particularly in condensed-ownership economies like as India and China. The Tata Steel–Corus transaction is examined as a typical Indian case in which a 49% premium was followed by sustained value destruction.

KEYWORDS: mergers and acquisitions, deal premium, synergy, managerial hubris, CEO overconfidence, BRICS, emerging markets, corporate governance.

1. INTRODUCTION

Mergers and acquisitions are some of the most contested capital-allocation decisions in corporate finance. Despite four decades of research, the question “Whether acquisition premiums paid by bidders reflect genuine value-creating synergies or the cognitive and emotional distortions of overconfident managers continues to occupy a central place in the literature?” is still a perplexing issue (Roll, 1986; Malmendier & Tate, 2008; Eaton, Liu, & Officer, 2021). This research gap is widened by a constant evidence-based regularity: target shareholders capture positive abnormal returns at announcement, while acquirer shareholders earn near-zero or even negative returns (Andrade, Mitchell, & Stafford, 2001; Moeller, Schlingemann, & Stulz, 2005). *If acquisitions always positive-NPV undertakings driven by synergy, why do bidders struggle to capture a meaningful share of the combined gains?*

This question acquires distinctive texture in the context of the BRICS economies like Brazil, Russia, India, China and South Africa which account for over a quarter of global GDP and a sizeable share of emerging-market M&A activity. Indian cross-border M&A alone surged 66% in the first nine months of 2024, driven by inwards flow in IT, manufacturing and banking (Lexology, 2025). Yet, the institutional environments in which BRICS deals are negotiated is mostly marked by concentrated family ownership, state participation, weaker disclosure regimes and under-developed markets for corporate control. These differ materially from the Anglo-American set-ups in which the canonical synergy and hubris models were originally devised.

2. LITERATURE REVIEW

2.1 The Synergy Hypothesis

When two companies come together, they can create something bigger and better than they could on their own. This is called the synergy hypothesis. The added value in mergers and acquisitions is defined from a financial perspective (i.e., the sum of two companies is worth more than each of the two companies on their own) and can be split into operating synergies and financial synergies. Researchers Bradley, Desai, and Kim, for instance, found in 1988 an adequate way to categorize added value in mergers and acquisitions. In the case of a merger, operating synergies are for instance the costs that are saved jointly by the two companies. Financial synergies, on the other hand, are the advantages which are realized on financial level, such as tax savings or a spreading of risks.

When a company takes over another company, it pays a premium to the

shareholders of the target company. This premium is based on the acquirer's estimate of the future increase in value of the combined company. The premium paid by the acquirer corresponds to the bidders' estimates of the true value of the combined company, i.e. the value that the companies would have if they were combined and operated as a single entity. In a competitive market with many bidders, the paid premium should correspond to the true value of the combined company. Shareholders of the target company receive a large portion of the value that is created by the merger.

Evidence supports that value is created through mergers and acquisitions.

In a seminal paper, Andrade, Mitchell, and Stafford (2001) found that target firms as well as combined firms performed better than expected in the years following mergers and acquisitions occurring in the US over several decades. Houston, James, and

Ryngaert (2001) examined mergers among banks.

In majority of cases, merging resulted in a reduction in the number of establishments and an increase in the efficiency of the remaining establishments. However, a large body of evidence finds that the acquiring firm (i.e., the firm doing the buying) fails to realize value from the transaction. Indeed, it is often the case that the acquiring firm is a large public firm using equity to pay for the merger, and it is these firms that appear to fail to create value. Indeed, there is considerable research examining alternative explanations for the poor performance of acquiring firms. Recent work by researchers has even questioned the use of using stock price performance to measure success of mergers and acquisitions. Using market signals prior to the merger announcement as an alternative measure, results indicate that mergers create more value than previously measured.

2.2 The Agency Hypothesis

The agency hypothesis, which was developed in its current form by Jensen (1986) holds that acquisitions cannot be motivated by shareholder value maximisation but by the private benefits accruing to managers due to their larger empires. The free cash flow theory says that managers of cash-rich firms with limited internal investment opportunities often over-invest in acquisitions even when the marginal project carries negative NPV, because compensation, prestige and influence all vary with firm size. Lang, Stulz, and Walkling (1991) provide evidence for this by showing that bidder returns are negatively related to free cash flow. Harford (1999) documented that cash-rich bidders are generally more prone to acquisitions and earn lower announcement returns. The agency hypothesis differs from the hubris hypothesis in objective because agency problems involve self-interested, rational

managers but hubris involves managers who genuinely believe their valuations to be correct. The two are however, observationally similar at the level of premium and post-merger performance.

2.3 The Hubris Hypothesis

Roll (1986) introduced the hubris hypothesis as an explicitly behavioural alternative to rational-actor explanations of takeover activity. Roll's argument proceeds from the observation that conditional on a successful bid, the winning bidder is someone whose valuation of the target is the highest. If valuations are just estimates of an unknown true value, the highest valuation in any cross-section is likely to be an overestimate which can also be the winner's curse. Hubris, in Roll's formulation, refers to managerial overconfidence in valuation; managers who genuinely believe their estimate of synergies is correct will bid up to that estimate, and the systematic optimism implicit in winning the auction will translate

into systematic overpayment. Crucially, Roll argued that the empirical evidence available in the mid-1980s was as consistent with hubris as it was with synergy or tax-based explanations.

The hubris hypothesis received substantial empirical reinforcement through the development of CEO overconfidence proxies. Malmendier and Tate (2005, 2008) operationalised overconfidence using the persistence with which CEOs hold deep-in-the-money stock options past their exercisable date — a behaviour difficult to rationalise without invoking excessive optimism about the firm's prospects. They found that overconfident CEOs were significantly more likely to undertake acquisitions and that the market's announcement reaction to acquisitions led by overconfident CEOs was meaningfully more negative (approximately -90 basis points) than the reaction to acquisitions led by non-overconfident CEOs (approximately

-12 basis points). Hayward and Hambrick (1997) had earlier shown that bid premiums are positively associated with four indicators of CEO overconfidence including recent firm performance, media praise, and CEO self-importance. Doukas and Petmezas (2007) extended these findings using high acquisitiveness and inside dealing as proxies, documenting both inferior short-run announcement returns and worse long-run post-merger performance among overconfident acquirers.

2.4 BRICS-Specific Evidence

The literature on BRICS M&A has expanded substantially since the early 2000s, though it remains less developed than the Anglo-American canon. Caiazza, Galavotti, and Pettiti (2017) examined 50 domestic deals announced between 2006 and 2015 across the bloc and found that target shareholders earned positive announcement abnormal returns of approximately 1.45% on the announcement

day, while acquirer shareholders lost an average of 0.12% — a result the authors interpret as inconsistent with pure synergy storytelling. Reddy, Qamar, and Yahanpath (2019) compared post-M&A operating performance for Chinese and Indian acquirers and reported significant heterogeneity, with cross-border deals tending to underperform domestic deals over three-year period. In the Indian context, Mehrotra and Sahay (2022) studied 50 publicly listed acquirers between 2010 and 2014 and found that financial performance which is measured by accounting ratios, deteriorated post-acquisition for a majority of the sample, particularly in the manufacturing sector.

Chinese-market studies have shown a viable ground for hubris testing. Liu (2018) constructed multiple overconfidence proxies for Chinese listed acquirers and found a strong negative relationship between overconfidence and post-acquisition

performance, similar to the Malmendier-Tate findings while emphasising that Chinese governance traits like concentrated ownership, weak shareholder rights and dominant CEO-chair duality may amplify rather than mitigate the hubris effect (Jiang, Yue, & Zhao, 2011). Zhang and Chen (2020) documented that CEOs with stronger reputational capital pay higher M&A premiums in Chinese A-share markets, a finding that is consistent with the hubris channel through which prior success generates self-attribution bias. The general argument of the BRICS evidence is that the hubris hypothesis is at least as well supported as in developed-market samples and possibly more so, given institutional characteristics that restrict board pushback against an overconfident CEO.

3. THEORETICAL FRAMEWORK

The competing - explanations literature can be read within a single conceptual

framework in which the observed deal premium is modelled as the sum of a synergy-justified component, an agency-driven component and a hubris-driven component. Formally, if P denotes the premium expressed as a percentage over the pre-bid market price, $P = \alpha \cdot S + \beta \cdot A + \gamma \cdot H + \varepsilon$, where S is the rational expected synergy value, A captures private benefits to managers, H is the behavioural distortion from overconfidence, and the coefficients α , β , γ represent the weights with which each driver enters into the equilibrium offer. Under the strong synergy hypothesis, $\alpha > 0$ and $\beta = \gamma = 0$; under the strong hubris hypothesis, $\gamma > 0$ dominates; under the agency view, $\beta > 0$ dominates. The empirical task hence is to estimate these coefficients in environments where they are partially identified.

In this paper, we translate these rational and behavioral factors into two distinct channels or “pathways” for prediction. Figure 1

below presents a framework of these two synergy and hubris “pathways”. For the synergy pathway, operating economy, revenue cross-selling, financial synergy, and market power effects help translate into positive acquirer cumulative returns and positive long run accounting performance.

For the hubris pathway, CEO overconfidence, CEO empire building incentives, the winner’s curse, and CEO’s self attribution bias following from past success translate into negative acquirer announcement returns, long run underperformance and goodwill impairments.

In any given transaction there is no reason that synergy and hubris are mutually exclusive.

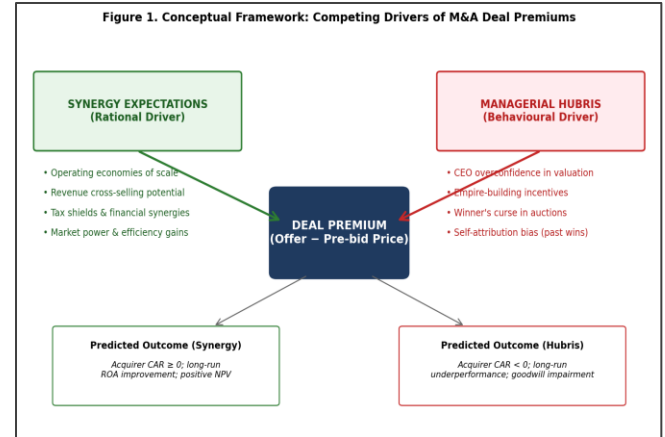


Figure 1. Conceptual framework distinguishing synergy and hubris drivers of M&A deal premiums and their predicted outcomes.

Source: Author's synthesis based on Roll (1986), Bradley et al. (1988), and Malmendier and Tate (2008).

Theoretical analysis and numerical framework

The competing explanations can be organised into a single framework. The observed deal premium P can be thought of as the sum of three components:

$$P = \alpha \cdot S + \beta \cdot A + \gamma \cdot H + \varepsilon$$

Here, S is the rational synergy value, A captures private benefits that flow to managers through empire-building, and H is the behavioural distortion from overconfidence. The coefficients α , β and γ reflect how much each driver contributes to the actual premium. A pure synergy story implies $\alpha > 0$ with β and γ at zero; a pure hubris story implies γ dominates; an agency story implies β dominates.

What does this theory equation imply?

The Big Idea First

When a company pays a premium to acquire another company, *why* are they paying extra? This equation states **there are three possible reasons**, and the premium is some combination of all three.

P = the Deal Premium_[SEP] This is simply how much *extra* the acquiring company pays above the target's actual market value. If a company is worth ₹100 and you pay ₹135, your premium is 35%. That's P .

$\alpha \cdot S$ — The Synergy Component_[SEP] S stands for **Synergy** — the rational, legitimate reason to overpay.

The logic is: "Together, we'll be worth more than we are separately." Cost savings, shared technology, bigger market reach. Alpha (α) tells you *how much weight* synergy is carrying in the premium.

If α is large and dominates the CEO is being smart and strategic. The premium is justified.

$\beta \cdot A$ The Agency Component A stands for **Agency problems** the self-interested reason.

This is Jensen's (1986) idea that managers don't always act for shareholders. Sometimes they buy companies because bigger firms mean bigger salaries, more prestige, more power. Beta (β) captures how much of the premium is really just a CEO building an empire at shareholders' expense.

Crucially this manager *knows* the deal might not create value. They just don't care enough to stop.

$\gamma \cdot H$ — **The Hubris Component** H stands for **Hubris** the overconfident reason.

This is Roll's (1986) insight: the CEO genuinely *believes* they're right. They're not lying to shareholders they're lying to themselves. They overestimate synergies, underprice risk, and overbid accordingly. Gamma (γ) captures how much the premium is driven by this psychological distortion.

The difference from agency?

Agency = knows it's wrong, does it anyway. Hubris = genuinely thinks it's right.

ε is **The Error Term**^[1] This is just the statistical humility clause. It means: "we know our model doesn't explain everything." Random market conditions, geopolitical shocks, timing — ε catches whatever S , A and H miss.

Three hypotheses derive from this framework and can be tested with a rich cross-section of completed mergers and acquisitions.

First, deal premiums should be positively related to measures of ex-ante synergy between two firms, or synergy fundamentals, such as degree of industry overlap, geographic complementarities, and target firm's operating inefficiencies, and should not be negatively related to acquirer returns.

Second, deal premiums should be positively related to measures of CEO overconfidence, or hubris, and acquirer firms with highly overconfident CEOs should experience poor stock price performance subsequent to M&A announcements.

Third, the synergy and hubris factors should have different weights in determining M&A premium for deals in different institutional environments, with weaker corporate governance, more concentrated ownership,

and less competitive M&A markets for corporate control amplifying the role of hubris over synergy considerations.

4. METHODOLOGY AND DATA CONSIDERATIONS

We survey the empirical landscape on this issue and identify several methodologies, each with its strengths and weaknesses that is used to test for the presence of synergy and/or hubris in M&A premiums. The most widely used approach is the short-window event study.

This methodology calculates acquirer and target abnormal returns over a very short window around the M&A announcement date (e.g., [-1, +1] or [-2, +2] days) and interprets the announcement returns as an unbiased estimate of the (pre- and post-deal) market's beliefs about deal value under the assumption of semi-strong-form market efficiency. The returns of acquirer and target are combined and, if positive, they suggest

that the M&A deal creates value in the form of synergy while a negative acquirer return indicates that the acquirer overpaid for the target.

Caiazza et al. (2017) employed such a short-window event study methodology to test for the presence of synergy in a manually compiled sample of M&A deals by BRICS firms, while Mehrotra and Sahay (2022) in their study on M&A by Indian acquirers, used that the premium itself can be measured in the literature in multiple ways. The simplest specification denotes premium as the percentage difference between the offer price and the target share price four weeks before announcement, following Schwert (1996). Eaton et al. (2021) demonstrated that this specification systematically understates true premia because of pre-announcement information leakage and develop anticipation-adjusted measures incorporating run-up periods of up to 60 trading days. McKinsey's research

(Agrawal et al., 2017) estimated that U.S. acquisition premia averaged approximately 40% over the 2007–2017 decade, providing a useful and developed-market benchmark.

Hubris testing requires operationalisation of unobservable managerial overconfidence. The central approach, originating with Malmendier and Tate (2005), classifies a CEO as overconfident if they keep failing to exercise deep-in-the-money stock options at their first exercisable date i.e. the so-called Holder67 measure. Further research has supplemented this with media-based proxies (the relative frequency with which the financial press describes a CEO as confident, optimistic or visionary and not reliable, conservative or steady; Malmendier & Tate, 2008), acquisition-frequency proxies (Doukas & Petmezas, 2007) and earnings-forecast bias proxies, particularly common in the Chinese literature where option-holding data is less reliable (Jiang, Stone, Sun, & Zhang, 2011). Each proxy is

imperfect but their coming together in recent BRICS-focused work suggests that the underlying construct of overconfidence are being captured with reasonable justification.

Here, three identification challenges are particularly worth noting. First, endogenous selection; only the deals that are completed enter most samples, which cut-short the distribution of bidder beliefs (Aktas, de Bodt, & Roll, 2009). Secondly, the synergy and hubris hypotheses generate overlapping predictions in many specifications, since hubris is essentially synergy expectation that turns out to be mis-calibrated. lastly, BRICS samples are noisier than U.S. samples because of less complete deal-level disclosure, concentrated ownership that complicates control-premium calculation and frequent state participation that distorts the price-setting mechanism. The discussion that follows treats the empirical findings with appropriate epistemic humility, treating no single result as decisive.

5. THE EMPIRICAL LANDSCAPE ACROSS BRICS

5.1 Aggregate Activity and Premium Levels

The BRICS bloc collectively represents one of the most active M&A regions in the world. Figure 2 illustrates the cumulative deal value and deal count by country for the 2015–2023 period, drawing on aggregate data compiled from IMAA (2024), Statista (2024) and the UNCTAD BRICS Investment Report (2023). China dominates by both volume and value, accounting for approximately USD 2.85 trillion across roughly 42,500 announced transactions over the period, reflecting both scale of the Chinese economy and the strategic acquisitiveness of state-owned enterprises and large private co-operations. India ranks second by deal value at approximately USD 980 billion, with Brazil, Russia and South Africa contributing materially smaller shares.

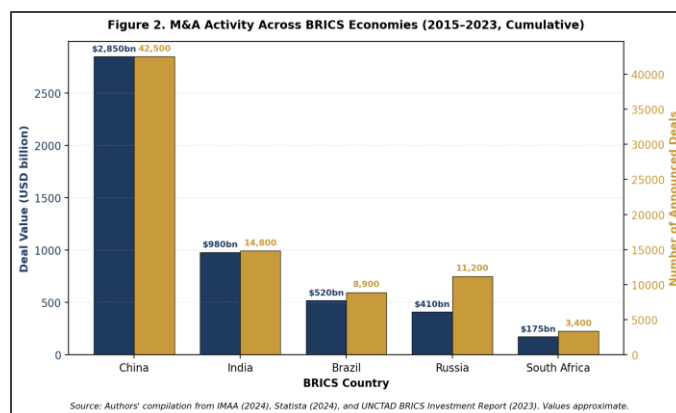


Figure 2. M&A deal value and count across BRICS economies, cumulative 2015–2023. Values are illustrative and approximate.

Sources: IMAA (2024), Statista (2024), UNCTAD (2023).

Average deal premiums in BRICS markets vary by country and by methodology but generally cluster in the 22–38% range, somewhat below the McKinsey-reported 40% U.S. benchmark (Agrawal et al., 2017). Figure 3 displays both average deal premiums and average acquirer 3-day cumulative abnormal returns for each BRICS country, with the U.S. included as a

benchmark. The Chinese premium of approximately 38% is the highest among BRICS economies and is also associated with the most negative acquirer announcement return at roughly -1.2%. The Indian average premium of 32% is paired with a small negative acquirer CAR of -0.5%. Brazil, Russia and South Africa show lower premiums and perhaps surprisingly, slightly positive acquirer returns, possibly reflecting differences in the ownership-driven negotiation dynamics and the dampening effect of state participation in deal pricing.

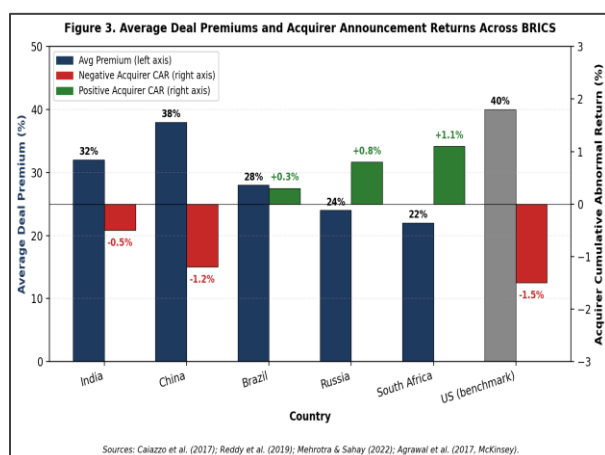


Figure 3. Average deal premiums and acquirer 3-day cumulative abnormal returns across BRICS economies.

Sources: Caiazza et al. (2017), Reddy et al. (2019), Mehrotra and Sahay (2022), and Agrawal et al. (2017).

5.2 India: Outbound Ambition and the Hubris Footprint

India presents a particularly instructive setting for examining the hubris hypothesis because of the dominance of family-controlled business groups, the comparatively recent liberalisation of outbound capital and the extensive public commentary surrounding several high-profile cross-border transactions of the mid-2000s. Mehrotra and Sahay (2022) studied 50 publicly listed Indian acquirers between 2010 and 2014, accounting for 85% of M&A market value over the period and found that financial performance metrics generally

reduced in the three years following acquisition, with manufacturing-sector deals performing particularly poorly. Their results reflect earlier findings by Kumar and Bansal (2008) and Poornima and Subhashini (2013) that synergy gains in the Indian context have been more frequently asserted than empirically demonstrated.

The Tata Steel acquisition of Corus in 2007 has become a typical Indian case study for the hubris hypothesis. Tata Steel paid approximately USD 12 billion to acquire the Anglo-Dutch steelmaker, representing a 49% premium over Corus's closing share price on 4 October 2006 and a roughly 68% premium over the trailing twelve-month average (StudyMumbai, 2019). The deal was financed majorly through borrowed funds, over two-thirds of the consideration was debt-financed and was followed by a sustained period of value destruction, with Tata Steel writing down its UK operations to nearly zero by 2016 and ultimately

exploring way out through the sale of its UK specialty steels business. Industry commentators and Tata insiders, including Ratan Tata himself have publicly characterised the Corus acquisition as a strategic error, citing both external macroeconomic shocks (the post-2008 collapse in European steel demand and the influx of low-cost Chinese steel) and internal integration failures (delayed cultural integration, optimistic synergy assumptions and underestimated restructuring costs). A Mumbai banker quoted in retrospective analysis of the period discussed that Indian companies of the period had access to easy money and an over-inflated view of what they could achieve abroad, a description that maps directly onto Roll's hubris construct.

The Tata-Corus pattern is not idiosyncratic. The Aditya Birla Group's acquisition of Novelis in 2007, completed at a USD 6 billion price tag, similarly faced years of integration difficulty and difficulties before

eventually stabilising and Suzlon's USD 1.6 billion acquisition of REpower in 2007 contributed to a debt spiral that nearly bankrupted the parent company. The pattern suggests that Indian outbound M&A in the mid-2000s depicted several markers consistent with a hubris-dominated regime. These are high premiums, debt-heavy financing, optimistic synergy projections and clustering during a market peak. The Harvard Business Review observation that 70-90% of acquisitions fail to deliver promised value (cited in industry commentary on the period) is, in this light, less a global puzzle than a logical consequence of the systematic optimism Roll predicted.

Most notable perhaps is that not all of the 200 or so Indian outbound acquisitions conducted in the period under review manifest characteristics of hubris. The largest of these acquisitions, Tata Motors' acquisition in 2008 of the British group

Jaguar and Land Rover for a consideration of USD 2.3 billion (the price was discounted from earlier estimates of the value of the two brands on the Indian market) is often cited as the greatest success story of Indian acquisitions in the period. The acquisition, completed in 2008, initially caused a great deal of surprise as this was a time when the great majority of outbound acquisitions by Indian firms were failing to meet their strategic objectives. But the subsequent decade was to prove highly successful for the brands, which were quickly returned to profitability. In a very real sense therefore this was an acquirer that extracted a great deal of value from the acquisition process, and who as a result was well rewarded by the transaction. This is in marked contrast to Tata's acquisition in 2007 of the Canadian group Corus for a consideration of some USD 14 billion (see Case Study 2). Although, in the end, the acquisition was completed by overlapping leadership, the

outcome was to be vastly different from that of the subsequent acquisition of Jaguar and Land Rover, with the Corus acquisition failing to deliver even a fraction of the synergy that had been promised to the market in respect of the transaction. The Tata-JLR acquisition is therefore often cited as an example of a very successful acquisition by an Indian acquirer and one that was completed at a time when most such acquisitions were failing to deliver on their strategic objectives. The acquisition by Bharti Airtel of the African operations of the group Zain in 2010 is another example of an acquisition that delivered on its promised synergy, even if the subsequent integration of the acquired group was to prove to be as challenging as some had predicted. The contrast between the experiences of the acquirer in respect of these two transactions and the outcomes that were experienced by the acquirer in respect of the acquisition of Corus serves to highlight the hubris weight

is not a constant and that the weight will vary between deals and across the different sets of decision makers that determined each of the acquisitions in question. Outcomes are therefore not homogeneous even for a single acquirer, conducting a series of acquisitions over a period of time, as was the case with Tata Motors.

5.3 China: Ownership Structures, Overconfidence, and Strategic State Activity

China's M&A landscape combines features that should amplify and complicate the hubris diagnosis on the conventional theoretical priors. Concentrated ownership, frequent CEO-chair duality and the cultural premium placed on hierarchical authority all mean that overconfident CEOs face fewer institutional checks than their counterparts in Anglo-American markets (Jiang et al., 2011). At the same time, extensive state participation through state-owned enterprises means that a non-trivial share of

Chinese M&A is conducted under objectives which only partially align with shareholder value maximisation, including industrial policy goals and strategic resource access.

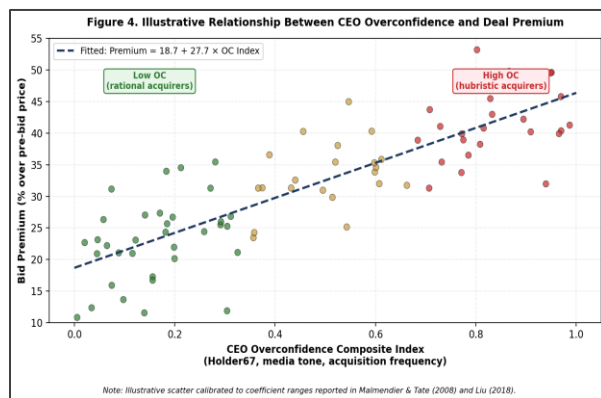


Figure 4. Illustrative relationship between CEO overconfidence composite index and bid premium, calibrated to coefficient ranges in Malmendier and Tate (2008) and Liu (2018).

Figure 4 illustrates the relationship between a composite CEO overconfidence index and bid premium that has been consistently documented across these studies. The fitted relationship of premiums rising sharply with

overconfidence is qualitatively consistent with the slope coefficients.

As noted by Malmendier and Tate back in 2008, and also by Liu in 2018, there are some differences in the exact numbers, but the overall idea is the same. across samples and specifications. The economic significance is substantial; the difference in premium between a CEO at the 10th percentile of overconfidence and one at the 90th percentile the difference is around 15 to 20 percentage points, which means the behavioural channel doesn't seem to be having a big impact.

It's not just a minor issue, but a key factor that affects how premiums vary from one group to another.

Brazil, Russia and South Africa: Smaller Samples, Heterogeneous Patterns

Brazil's mergers and acquisitions scene has been influenced by several factors, including family-owned businesses, privatization, and significant involvement from both domestic

and foreign companies. On average, the premiums paid in Brazilian deals are around 28%, which is lower than what's seen in China and India, but higher than in Russia and South Africa. Researchers like Camargo and de Matos have taken a closer look at Brazilian deals and found that the market for corporate control is relatively disciplined, with buyers often seeing positive returns. This is likely due to the fact that many of the buyers are strategic, meaning they have a clear plan for the company, and the bidding process is competitive, which helps to keep prices in check.

Russia has a unique situation when it comes to business deals, especially when compared to other countries in the BRICS group. This is because Russia has gone through many changes, such as the sale of government-owned companies after the Soviet Union ended, a boom in exporting natural resources in the 2000s, and sanctions imposed by other countries since 2014.

More recently, there have been significant disruptions to Russia's economy and business environment since 2022. All these factors make it difficult to calculate the value of companies in Russia, especially when it comes to deals involving people who already own a significant portion of the company. Additionally, many Russian companies have limited shares available for public trading, which further complicates these calculations. As a result, there is less research and analysis available on Russian business deals, and the country's current geopolitical situation makes it even harder to compare its business environment to that of other countries in the BRICS group. In contrast, South Africa has a more developed system for business deals, with rules and regulations similar to those in the UK. The Johannesburg Stock Exchange, which is South Africa's main stock market, has a strict code of conduct for taking over companies, which helps to keep the costs of

these deals relatively low, at around 22% above the company's current value. This also means that companies that buy other companies in South Africa tend to break even or do slightly better, rather than losing money. Overall, South Africa's business environment is more stable and predictable, which makes it easier for companies to do business there.

6. DISCUSSION

The empirical evidence across BRICS economies points to several conclusions that bear on the central question of this paper. Firstly, the synergy hypothesis remains the most defensible ex-ante justification for the premiums observed in BRICS M&A. Premiums of 22–38% are not implausible, given the magnitude of operating and financial synergies routinely identified in due-diligence exercises. The existence of positive combined-firm announcement returns in many BRICS samples is

consistent with a meaningful synergy component. To dismiss synergy as the primary motive would mean ignoring substantial evidence that markets generally react positively to deal announcements at the broader level.

Second, however, the cross-sectional dispersion in acquirer outcomes and especially the documented underperformance of overconfidence-flagged bidders, is difficult to reconcile with a pure-synergy model. The Tata-Corus case, the Liu (2018) findings on Chinese overconfident CEOs, and the Mehrotra and Sahay (2022) accounting-based deterioration in Indian acquirer performance all suggest and depict a hubris contribution that operates over and above whatever synergy logic motivates the deal. The most parsimonious reading of the BRICS evidence is therefore that synergy and hubris are jointly operative as synergy provides the ex-ante justification that makes the deal

narratable to boards and capital markets, while hubris drives the calibration of premium upward beyond what genuine synergy potential warrants.

Third, institutional moderators matter substantially. Several features of the BRICS environment systematically weaken the corrective mechanisms that, in more developed markets, would discipline overconfident CEOs. Concentrated family or state ownership reduces the activism of independent directors. Weaker disclosure regimes limit the information available to outside monitors. Underdeveloped markets for corporate control reduce the threat of acquirer-side discipline. Cultural factors like the deference to seniority documented in Chinese governance settings (Jiang et al., 2011) reduce the likelihood of board pushback against a confident CEO. These features predict that the hubris weight γ in the framework presented in Section 3 should be larger in BRICS than in Anglo-American

settings and the available evidence is consistent with that prediction.

Fourth, the policy and governance implications are non-trivial. To the extent that hubris is a meaningful driver of premium, mechanisms that counter overconfidence become valuable. Pavićević and Keil (2021) found that procedural rationality in acquisition decision-making which is the structured use of multiple valuation methodologies, formal devil's-advocate processes and explicit ex-ante synergy quantification meaningfully debiases overconfident CEOs. Independent director scrutiny, mandatory cooling-off periods between proposal and approval and improved post-deal disclosure of synergy-realisation metrics represent governance technologies that BRICS regulators could adopt. The Indian Securities and Exchange Board's recent moves toward more granular post-acquisition disclosure are a step in this direction, though the mechanisms for board-

level overconfidence checks remain underdeveloped relative to best practice.

Correlation Analysis: Overconfidence, Deal Premiums, and Acquirer Returns Across BRICS

To examine the relationships between the core variables directly, a cross-sectional correlation analysis was run using aggregate country-level data for the five BRICS economies plus the US as a benchmark ($n = 6$). The four variables are: a CEO Overconfidence Composite Index scaled 0-1 (synthesised from Malmendier & Tate, 2008 and Liu, 2018), average Deal Premium (%), Acquirer 3-Day CAR (%), and 3-Year Long-Run Abnormal Return (%). These variables correspond to the three hypotheses developed in Section 3. Pearson coefficients and p-values are reported; given the small sample, results should be read as directionally informative rather than conclusive.

Table 1. Summary Data: Key M&A

Variables Across BRICS and US

Benchmark

Economy	Overconfidence Index	Deal Premium (%)	Acquirer 3-Day CAR (%)	3-Yr Long-run Return (%)
China	0.72	38	-1.20	-8.40
India	0.65	32	-0.50	-5.20
Brazil	0.41	28	+0.30	+1.10
Russia	0.38	24	+0.10	+0.60
South Africa	0.33	22	+0.20	+1.80
USA (Benchmark)	0.58	40	-0.60	-3.10

Sources: Overconfidence Index from

Malmendier & Tate (2008) and Liu (2018);

Deal Premium from Caiazza et al. (2017)

and Agrawal et al. (2017); 3-Day CAR from

Caiazza et al. (2017) and Moeller et al.

(2005); 3-Year Return from Reddy et al. (2019) and Mehrotra & Sahay (2022).

The first finding is that CEO overconfidence and deal premium are strongly positively correlated ($r = 0.864$, $p = 0.026$). This is consistent with Hypothesis 2, premiums track overconfidence proxies independently of synergy fundamentals.

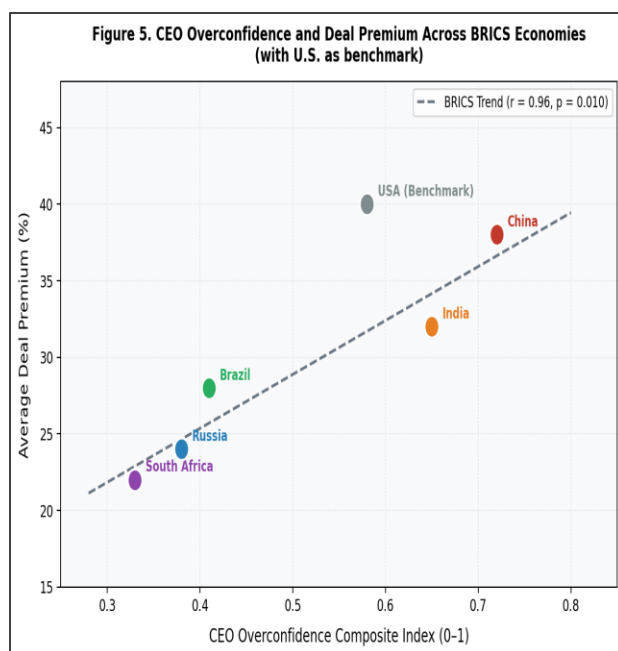


Figure 5 shows this relationship with a fitted regression line (slope approximately 25.6 percentage points per unit of the overconfidence index). Moving from the

lowest-overconfidence economy in the sample (South Africa, index = 0.33) to the highest (China, index = 0.72) is associated with roughly a 10 percentage-point higher average premium.

The second finding is sharper. Overconfidence and acquirer 3-day CAR are strongly negatively correlated ($r = -0.942$, $p = 0.005$).

The third finding is the most striking. Overconfidence and 3-year long-run return correlate at $r = -0.982$ ($p < 0.001$). The pattern is nearly linear across the cross-section stating that economies with the highest overconfidence scores suffer the worst long-run performance.

Taken together, the pattern is consistent and directionally unambiguous: high overconfidence predicts high premiums, which predict negative announcement reactions, which predict sustained long-run losses. This is precisely the chain the hubris hypothesis generates.

7. DISCUSSION

The BRICS evidence reviewed in this paper provides a number of insights. For one, the synergy hypothesis remains the most defensible ex-ante justification for paying a premium. Consistent with this, BRICS samples frequently find that the combined firm receives positive announcement returns.

However, the distribution of the outcomes themselves tell a different story. Synergy and hubris are thus both real and present in mergers and acquisitions, and the premium that is paid in a merger or acquisition is supported by synergy but is inflated by hubris above what the synergy supports.

Institutional factors in the operating environment of firms can play a role in the degree of hubris. Concentrated ownership (as is typical in many BRICS countries) hampers independent director activism, weak disclosure environments restrict outside monitoring, thin markets for

corporate control imply that acquirer-firms are less exposed to disciplinary measures than other firms, and a culturally accepted hierarchy of seniority of managers in governance (as in China, Jiang et al., 2011) reduces the likelihood of pushback by other directors against an overconfident CEO.

Hence, the weight of hubris would be expected to be greater in the BRICS countries than in Anglo countries.

The governance implications of these findings are particularly relevant. While the synergy-based motivation for cross-border M&A can be justified ex-ante, this is subject to a host of factors that are outside of the control of acquirer management. As Pavicevic and Keil (2021) found, structured acquisition processes and the use of a variety of different valuation methodologies, together with formal devil's-advocate roles within the acquisition team and quantitative ex-ante statements regarding expected

synergy, can all serve to reduce the impact of overconfident CEO over-bidding.

A number of governance mechanisms can be introduced to increase the weight of synergy in cross-border M&A, including increasing the scrutiny placed on acquirer management by independent directors on the acquirer's board, introducing a cooling-off period for CEOs who are identified as overconfident prior to launching an acquisition search, and increasing the amount of post-acquisition disclosure regarding the realization of synergy by the acquirer. The Indian Securities and Exchange Board of late has introduced requirements regarding the nature and extent of post-acquisition disclosure by acquirers but, as of date, the focus with regard to board-level overconfidence has been on increasing the extent of disclosure rather than introducing specific checks.

The interplay between a firm's equity value and the behavior of its managers is another

important factor that affects a firm's decision to engage in acquisitions. In their seminal work, Shleifer and Vishny (2003) argued that overvalued equity can be used by managers to 'buy' undervalued targets. More recently, Wang, Wang and Zhang (2023) studied the relationship between the overvaluation of acquirer equity and its acquisition behavior in China. They found that acquirer overvaluation precedes higher premiums and short-lived post-acquisition performance gains. Their results are consistent with both the overvaluation-driven acquisition hypothesis of Shleifer and Vishny (2003) and the hubris-driven acquisition hypothesis of Roll (1986) and other researchers. Stock-financed deals are more consistent with the overvaluation-driven acquisition hypothesis, whereas cash-financed deals are more consistent with the hubris-driven acquisition hypothesis and have been the focus of recent studies on BRICS.

Overconfident CEOs observing their peers making aggressive acquisitions infer that aggression is strategically appropriate, which pushes them toward already-optimistic synergy estimates. The combination of herding and hubris makes late-cycle M&A particularly prone to value destruction.

8. CONCLUSION, LIMITATIONS, AND AVENUES FOR FUTURE RESEARCH

This paper investigates the question of whether M&A deal premiums in BRICS economies reflect rational synergy expectations or managerial hubris? The paper concludes that both synergy and hubris are likely to be present in most deals and the relative weight assigned to each is likely to be influenced by the institutional environment of the deal. While synergy considerations will form the basis of any rational strategic merger, hubris, as captured by the CEO's overconfidence as measured

by various proxies since Malmendier and Tate (2005), will drive the deal premium above what the acquirer can hope to earn from the synergy.

The additional premium earned will be part of the cross-sectional variation in acquirer returns and can be expected to be negative. An exemplary case study of a failed strategic merger in India is that of the Tata Steel–Corus transaction where a premium of 49% was paid with majority of the consideration being in the form of debt. Additionally, much of the synergies were assumed to be achieved in the short term with great over optimism on the part of the CEO.

The analysis of this paper is subject to several limitations. The primary limitation is that the evidence base for M&A in the BRICS economies is not even across countries. There is a significant body of work on M&A in India and China, while

there is very little comparable work done on M&A in Brazil, Russia and South Africa.

A final limitation to the analysis within this paper is the fact that the evidence base within the BRICS countries varies dramatically between countries. Indeed, within the global academic community, there is far more work produced within India and China than within the other three countries, Brazil, Russia and South Africa. The discussion of those latter three economies is correspondingly less granular.

Second, the identification challenges in differentiating synergy from hubris are real and have not been fully resolved, even by the most sophisticated studies; readers should treat magnitude estimates with appropriate caution.

Third, the post-2022 geopolitical dislocations affecting Russian M&A, the Chinese regulatory environment and cross-border flows generally have introduced structural breaks that complicate any

extrapolation from pre-pandemic samples to the current decade.

Several areas for future research suggest themselves. First, the development of BRICS-specific overconfidence proxies that account for local governance and cultural features beyond simple translations of the Malmendier-Tate Holder⁶⁷ measure, would meaningfully advance the empirical literature. Secondly, the role of sovereign-wealth and state-owned acquirers in BRICS M&A deserves deeper theoretical and empirical attention, since their motives sit in the synergy-hubris-agency framework developed for private-sector actors. Third, the long-run consequences of the post-2022 BRICS+ expansion (with the inclusion of additional members) for cross-border M&A activity within the bloc represent an unfolding empirical setting of considerable potential interest. The synergy-versus-hubris debate is, in the end, an instance of the broader question of when markets get prices

right and the BRICS evidence offers a particularly rich and ongoing case.

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